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EXNO:5

DATE:

TITLE:

ATM System – UML Sequence Diagram for Withdrawal Transaction and print receipt

AIM:

To design and represent the dynamic interaction of classes in an ATM system using a UML sequence diagram.

SYSTEM OVERVIEW:

The ATM Withdrawal System is designed to allow customers to securely withdraw money from their bank account using an Automated Teller Machine (ATM). The system connects the customer, ATM machine, and bank server to complete the transaction safely and efficiently.

When a customer inserts the ATM card and enters the correct PIN, the ATM verifies the details with the bank server. After successful authentication, the customer can select the withdrawal option, enter the amount, and receive the cash if sufficient balance is available. The system then updates the account balance and records the transaction.

This system ensures security, accuracy, and reliability during the withdrawal process.

CLASS DIAGRAM:

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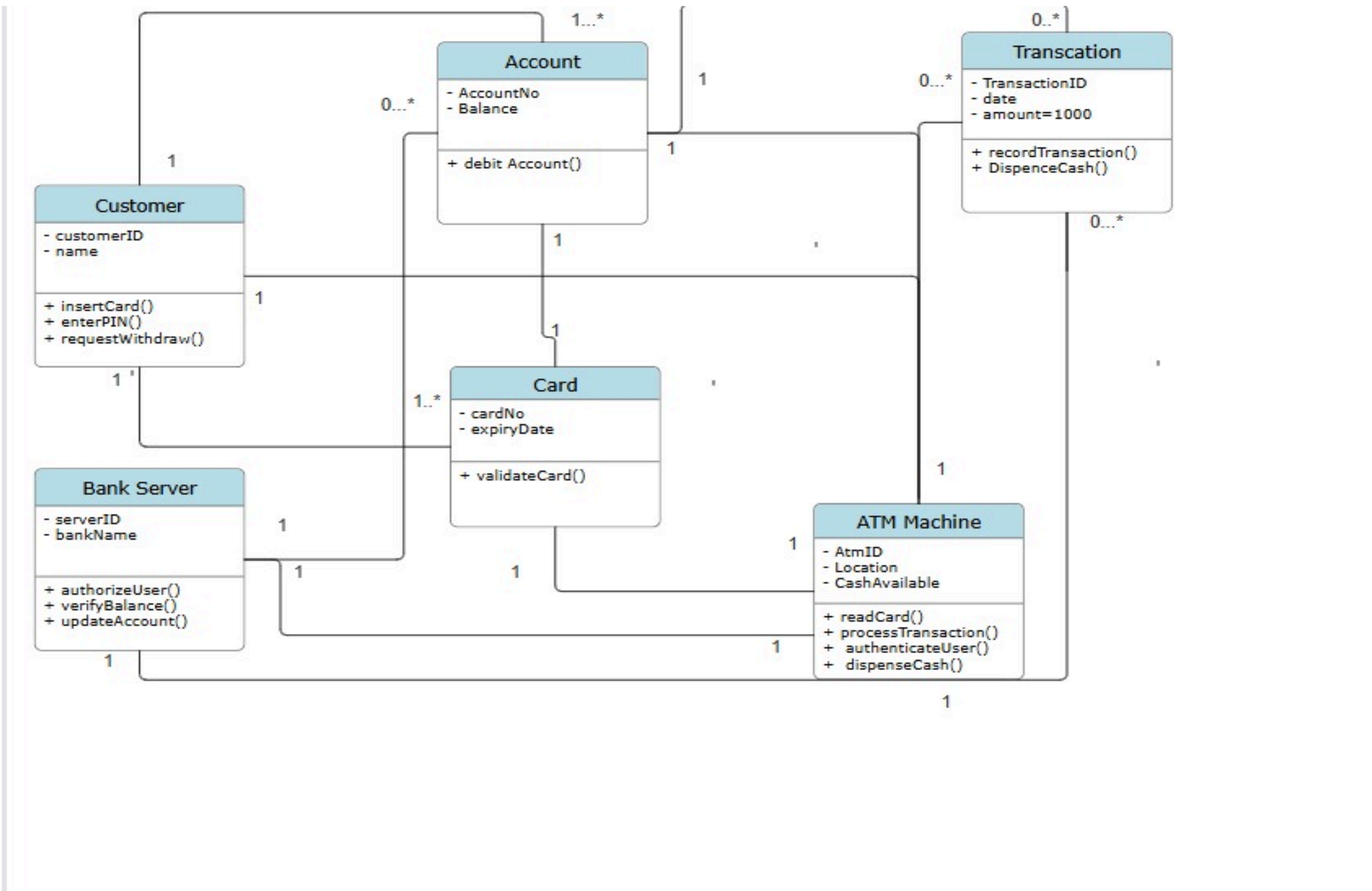

The Class Diagram of the ATM Withdrawal System represents the static structure of the system. The main classes involved are:

- Customer – Contains customer details such as name, account number, and PIN.
- ATM Machine – Handles user input, displays options, and dispenses cash.
- Bank Server – Verifies account details and processes transactions.
- Account – Stores account balance and performs debit operations.
- Transaction – Records transaction details like date, amount, and type.

  The relationships between these classes show how they interact.

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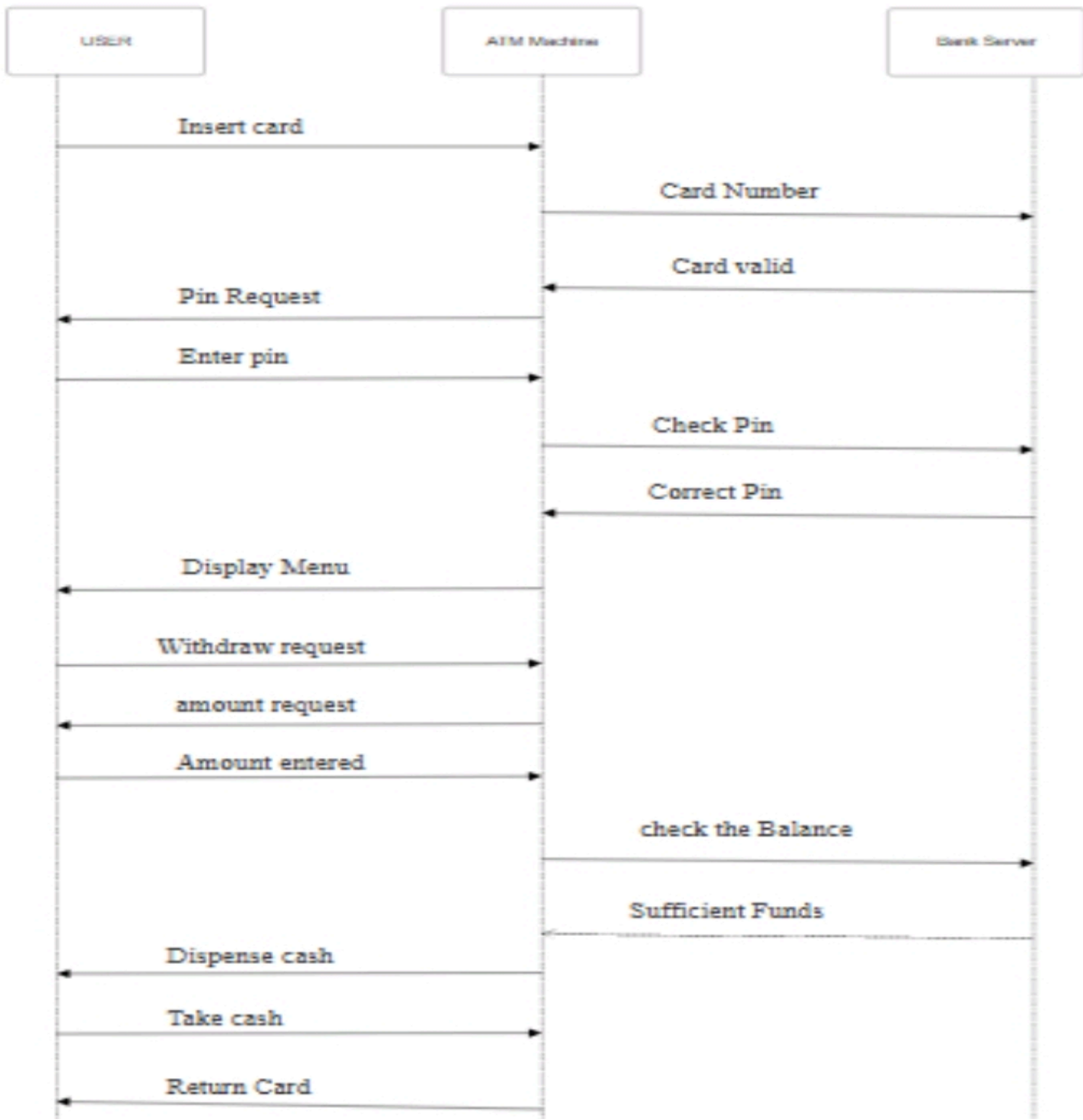
Time sequence diagram



The Time Sequence Diagram shows the step-by-step interaction during the ATM withdrawal process. The customer inserts the card and enters the PIN. The ATM sends the details to the bank server for verification. After successful authentication, the customer enters the withdrawal amount. The bank server checks the balance, approves the request, and the ATM dispenses the cash. Finally, the account balance is updated and the transaction is recorded

CONCLUSION:

The ATM Withdrawal System explains how UML diagrams help in designing a secure and structured system. The Class Diagram shows the system structure, and the Time Sequence Diagram shows how components interact during a transaction. This experiment highlights the importance of proper planning and modeling in software development.

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